

第七章 参数估计

$$2. (1) E(X) = \int_c^{+\infty} x f(x) dx = \int_c^{+\infty} \theta c^\theta x^{-\theta} dx = \frac{[x^{1-\theta}]_c^{+\infty}}{1-\theta} \theta c^\theta$$

$$\theta c = (\theta-1)\bar{x}$$

$$\theta(c-\bar{x}) = -\bar{x}$$

$$\hat{\theta} = \frac{-\bar{x}}{c-\bar{x}} = \frac{\bar{x}}{\bar{x}-c} = \frac{\frac{1}{n} \sum_{i=1}^n x_i}{\frac{1}{n} \sum_{i=1}^n x_i - c}$$

$$= x - \frac{c^{1-\theta}}{1-\theta} \theta c^\theta = \frac{\theta}{\theta-1} c = \bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$$

矩估计量

$$\hat{\theta} = \frac{\bar{x}}{\bar{x}-c}$$

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$$3. (1) L(x_1, \dots, x_n; \theta) = \prod_{i=1}^n P(X=x_i) = \prod_{i=1}^n \theta c^\theta x_i^{-(\theta+1)} = \theta^n c^{n\theta} \prod_{i=1}^n x_i^{-(\theta+1)}$$

$$\ln L = n \ln \theta + n\theta \ln c - (\theta+1) \ln \prod_{i=1}^n x_i$$

$$= n \ln \theta + n\theta \ln c - (\theta+1) \left[\sum_{i=1}^n \ln x_i \right]$$

$$\frac{d \ln L}{d \theta} = \frac{n}{\theta} + n \ln c - \sum_{i=1}^n \ln x_i = 0 \quad \hat{\theta} = \frac{n}{\sum_{i=1}^n \ln x_i - n \ln c}$$

最大似然估计值

最大似然估计量

$$\hat{\theta} = \frac{n}{\sum_{i=1}^n \ln x_i - n \ln c}$$

$$4. (1) E(X) = \sum_{k=1}^3 x_k p_k = \theta^2 + 4\theta(1-\theta) + 3(1-\theta)^2$$

$$= \theta^2 + 4\theta - 4\theta^2 + 3(1 - 2\theta + \theta^2)$$

$$= \theta^2 + 4\theta - 4\theta^2 + 3\theta^2 - 6\theta + 3$$

$$= -2\theta + 3 = \bar{x} = \frac{1+2+1}{3} = \frac{4}{3}$$

矩估计值为 $\theta = -\frac{1}{2}$

$$-2\theta = 2 \quad \theta = -\frac{1}{2}$$

$$2\theta = 3 - \frac{4}{3} = \frac{5}{3}$$

$$\theta = \frac{5}{6}$$

$$(2) L(x_1, \dots, x_n; \theta) = \prod_{k=1}^3 P(X=x_k) = \theta^2 \cdot 2\theta(1-\theta) \cdot \theta^2 = 2\theta^5(1-\theta)$$

$$\ln L = \ln 2 + 5 \ln \theta + \ln(1-\theta)$$

$$\frac{d(\ln L)}{d\theta} = \frac{5}{\theta} + \frac{1}{1-\theta} \cdot (-1) = \frac{5}{\theta} + \frac{1}{\theta-1} = 0$$

最大似然估计为 $\frac{5}{6}$

$$5(1-\theta) + \theta = 0$$

(2) λ 的最大似然估计量及矩估计量

$$6\theta = 5$$

$$\theta = \frac{5}{6}$$

$$\text{矩估计量} \quad E(X) = \lambda = \frac{1}{n} \sum_{i=1}^n x_i \quad \lambda = \bar{x}$$

$$\text{最大似然估计} \quad P(X=x_i) = \frac{\lambda^{x_i} e^{-\lambda}}{x_i!}$$

$$L(x_1, \dots, x_n; \theta) = \frac{\lambda^{\sum_{i=1}^n x_i} e^{-n\lambda}}{\prod_{i=1}^n x_i!}$$

$$\ln L = \sum_{i=1}^n x_i \ln \lambda - n\lambda - \sum_{i=1}^n \ln x_i$$

$$\frac{d(\ln L)}{d\lambda} = \frac{\sum_{i=1}^n x_i}{\lambda} - n = 0 \quad \lambda = \frac{1}{n} \sum_{i=1}^n x_i$$

最大似然估计量

$$\lambda = \bar{x}$$



$$10. E\left(c \sum_{i=1}^{n-1} (X_{i+1} - X_i)^2\right) = \sigma^2 = D(X)$$

$$\begin{aligned} c \sum_{i=1}^{n-1} (X_{i+1} - X_i)^2 &= c \sum_{i=1}^{n-1} (X_{i+1}^2 - 2X_{i+1}X_i + X_i^2) \\ &= c \left(\sum_{i=1}^{n-1} X_{i+1}^2 - 2 \sum_{i=1}^{n-1} X_{i+1}X_i + \sum_{i=1}^{n-1} X_i^2 \right) \end{aligned}$$

$$\sum_{i=1}^{n-1} X_{i+1}^2 = X_2^2 + \dots + X_n^2 \quad \sum_{i=1}^{n-1} X_{i+1}X_i = X_1X_2 + X_2X_3 + \dots + X_nX_{n-1}$$

$$\sum_{i=1}^{n-1} X_i^2 = X_1^2 + \dots + X_{n-1}^2$$

$$D(X) = \sum_{i=1}^n (X_i - \mu)^2 = \sum_{i=1}^n X_i^2 - 2 \sum_{i=1}^n X_i \mu + n \mu^2$$

$$D(X) = E(X^2) - [E(X)]^2 = \sum_{i=1}^n X_i^2 - n \bar{X}^2$$

$$\begin{aligned} E\left(c \sum_{i=1}^{n-1} (X_{i+1} - X_i)^2\right) &= c \sum_{i=1}^{n-1} \left\{ D(X_{i+1} - X_i) + [E(X_{i+1} - X_i)]^2 \right\} \\ &= c \sum_{i=1}^{n-1} \left\{ D(X_{i+1}) + D(X_i) + [E(X_{i+1}) - E(X_i)]^2 \right\} \\ &= c \sum_{i=1}^{n-1} [D(X_{i+1}) + D(X_i)] = c \cdot 2\sigma^2(n-1) = \sigma^2 \\ &\quad c = \frac{1}{2(n-1)} \end{aligned}$$

$$(2) E(\bar{X}^2 - c S^2) = \mu^2$$

$$E(\bar{X}^2 - c S^2) = E(\bar{X}^2) - c E(S^2)$$

$$= D(\bar{X}) + [E(\bar{X})]^2 - c \sigma^2 = \frac{\sigma^2}{n} + \mu^2 - c \sigma^2 = \mu^2$$

$$\begin{aligned} E(S^2) &= E\left[\frac{1}{n-1} \left(\sum_{i=1}^n X_i^2 - n \bar{X}^2\right)\right] = \frac{1}{n-1} E\left(\sum_{i=1}^n X_i^2\right) - \frac{n}{n-1} E(\bar{X}^2) \\ &= \frac{1}{n-1} \left[\sum_{i=1}^n [D(X_i) + [E(X_i)]^2] - n \{D(\bar{X}) + [E(\bar{X})]^2\} \right] \end{aligned}$$

$$\begin{aligned} D(\bar{X}) &= D\left(\frac{1}{n} \sum_{i=1}^n X_i\right) = \frac{1}{n^2} D\left(\sum_{i=1}^n X_i\right) = \frac{\sigma^2}{n} = \frac{1}{n-1} (n\sigma^2 + n\mu^2 - \sigma^2 - n\mu^2) \\ n D(\bar{X}) &= \sigma^2 \end{aligned}$$

$$E(\bar{X}) = E\left(\frac{1}{n} \sum_{i=1}^n X_i\right) = E(X) = \mu$$

$$12. E(T_1) = \frac{1}{6} [E(X_1) + E(X_2)] + \frac{1}{3} [E(X_3) + E(X_4)] = \theta$$

$$E(T_2) = \frac{1}{5} [E(X_1) + 2E(X_2) + 3E(X_3) + 4E(X_4)] = 2\theta$$

$$E(T_3) = \frac{1}{4} [E(X_1) + E(X_2) + E(X_3) + E(X_4)] = \theta$$

T_1, T_3 是 θ 的无偏估计量。

$$D(T_1) = \frac{1}{36} [D(X_1) + D(X_2)] + \frac{1}{9} [D(X_3) + D(X_4)] = \frac{1}{36} 2\theta^2 + \frac{1}{9} 2\theta^2 = \frac{2+8}{36} \theta^2 = \frac{10}{36} \theta^2$$

$$D(T_3) = \frac{1}{16} \sum_{i=1}^4 D(X_i) = \frac{4\theta^2}{16} = \frac{1}{4} \theta^2 \quad D(T_1) > D(T_3)$$

T_3 更有效。

